



eB Publications
2026 Sep 19th

**The Electricity
Foundation**

Understand the power.
Recognise the danger.
Know what to do.



UK ADULT ELECTRICITY LEVEL 1

A beginner's guide to understanding electricity,
UK homes, electrical safety and everyday
electrical language

Final Edition — Beginner Course



**UNDERSTAND
THE BASICS**



**RECOGNISE
THE RISKS**



**LEARN
UK TERMINOLOGY**



**MAKE SAFER
DECISIONS**

The Electricity Foundation
Knowledge today. Safer tomorrows.

UK ADULT ELECTRICITY - LEVEL 1 By eB Publications 2026 Sep 19th

The Electricity Foundation

Understand the power. Recognise the danger. Know what to do.

Final Edition — Beginner Course

01 — COURSE TITLE

UK Adult Electricity Level 1

The Electricity Foundation

**A beginner's guide to understanding electricity, UK homes, electrical safety
and everyday electrical language**

02 — SUBTITLE

**From “What is a volt?” to “When should I stop and call an electrician?” — explained
simply, accurately and practically.**

03 — COURSE OVERVIEW

Electricity is part of almost every modern home.

You switch on a light, charge a phone, boil a kettle or use a washing machine — but underneath
these simple actions is a system involving:

- Electrical charge
- Voltage
- Current
- Resistance
- Power
- Energy
- Circuits
- Conductors
- Insulation
- Protection
- Earthing
- Mains supply

This course turns those ideas into a **simple mental model**.

This course is designed for:

- Complete beginners
- Adults returning to education
- Homeowners and tenants
- People preparing for further electrical study
- People who simply want to understand electricity better

This course is not:

- An electrician qualification
- A substitute for practical training
- Permission to carry out electrical installation work
- A complete domestic electrical installation course

The Level 1 goal

By the end, you should be able to **understand electricity, recognise electrical hazards, understand common UK electrical terminology and make safer decisions around electrical equipment**.

04 — LEARNING OBJECTIVES

By completing this course, you should be able to:

-  **Understand** the basic nature of electricity.

- ✓ **Explain** the idea of an electrical circuit.
 - ✓ **Identify** voltage, current, resistance, power and energy.
 - ✓ **Explain** the difference between AC and DC.
 - ✓ **Identify** the basic roles of live, neutral and earth.
 - ✓ **Explain** the purpose of fuses, circuit breakers and RCDs.
 - ✓ **Recognise** common electrical hazards.
 - ✓ **Understand** basic UK domestic electrical terminology.
 - ✓ **Use** simple electrical formulas.
 - ✓ **Calculate** basic electrical values.
 - ✓ **Recognise** when a situation should be left to a competent/qualified professional.
-

05 — INDEX

FOUNDATION

1. Electricity: The Big Picture
2. Charge, Electrons and Electrical Energy
3. Circuits: The Complete Path

CORE ELECTRICAL IDEAS

4. Voltage
5. Current
6. Resistance
7. Ohm's Law
8. Power and Energy

UK ELECTRICITY

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10. AC and DC
11. Live, Neutral and Earth

- 12. UK Plugs, Sockets and Fuses
- 13. The Consumer Unit

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- 15. Common Electrical Hazards
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- 21. Key Takeaways
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 - 23. Q&A
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 - 25. Level 1 Test
 - 26. Answer Key
 - 27. Level 1 Completion Check
-

06 — COURSE GUIDELINES

The Proios Learning Route

Use:

Learn → Understand → Recall → Revise → Apply → Test → Improve

Don't try to memorise everything immediately.

First understand the **idea**.

Then learn the **word**.

Then practise using the idea.

Easy Glossary rule

Whenever you meet a difficult electrical word:

Stop → Find it in the Easy Glossary → Read the simple meaning → Continue.

The Easy Glossary deliberately uses very simple language while keeping the main technical meaning intact.

Golden Safety Rule

Level 1 knowledge does not equal electrical competence.

Never use this course as instructions for working on live mains electricity.

For electrical installation, alteration, testing or repair work beyond your competence, use an appropriately competent/qualified professional.

07 — INTRODUCTION

Electricity without the mystery

Electricity can appear complicated because electricians use many unfamiliar words.

But Level 1 electricity can be organised around a small number of questions:

1. What is moving?

Electric charge.

2. What makes it move?

Electrical potential difference — voltage.

3. How much is moving?

Current.

4. What makes movement harder?

Resistance.

5. What does the electricity do?

It can transfer energy into:

- Light
- Heat
- Movement
- Sound
- Computing
- Communication
- Other forms of useful energy

6. What keeps us safer?

Things such as:

- Insulation
- Earthing
- Fuses
- Circuit breakers
- RCDs
- Safe design
- Correct use
- Inspection and testing

That is the foundation.

08 — MAIN COURSE

MODULE 1 — ELECTRICITY: THE BIG PICTURE

Electricity is not one single thing.

It is a broad term used for phenomena involving **electric charge, electric fields, current and electrical energy**.

For Level 1, the most useful starting point is:

Electrical systems allow electrical energy to be transferred and used.

For example:

Electrical energy → lamp → light

Electrical energy → heater → heat

Electrical energy → motor → movement

Easy Glossary

Electricity: A group of effects involving electric charge and electrical energy.

Energy: The ability to cause change or do work.

Electrical system: Different parts working together to produce, control or use electricity.

MODULE 2 — CHARGE AND ELECTRONS

Matter contains tiny particles.

One important particle is the:

Electron

An electron has a **negative electric charge**.

In metals such as copper, electrons can move through the material.

This movement of charge is associated with **electric current**.

 **Important accuracy point**

It is useful for beginners to think:

“Electric current involves moving electrons.”

But electricity is **not simply a substance called “electricity” flowing through a wire like water.**

The water analogy helps explain some relationships, but it has limits.

Easy Glossary

Electron: A tiny particle with a negative electric charge.

Charge: A property that can make electrical forces happen.

Metal: A type of material in which electric charge can often move relatively easily.

MODULE 3 — CIRCUITS

A **circuit** provides a path for current.

A basic electrical circuit can contain:

Source → Conductors → Load → Return path

For example:

Battery → wire → lamp → wire → battery

If the required path is complete, the circuit can operate.

If the path is interrupted, the intended current cannot continue through that path.

Open and closed circuits

	State	Basic idea
	Closed	Complete path

Open

Broken/interrupted path

Memory trick

Closed = complete

Open = interrupted

The four basic parts

Source

Provides electrical energy or electrical potential.

Conductor

Provides a path for current.

Load

Uses electrical energy.

Switch

Controls whether a circuit path is connected or interrupted.

Easy Glossary

Circuit: A planned path that allows an electrical system to operate.

Source: Something that supplies electrical energy or electrical potential.

Load: Something that uses electrical energy.

Switch: A device that opens or closes a circuit path.

Conductor: A material that allows electric charge to move relatively easily.

MODULE 4 — VOLTAGE

What is voltage?

Voltage is **electrical potential difference**.

It is measured in:

volts (V)

A beginner-friendly description is:

Voltage is the electrical push or difference in electrical potential that can drive current.

Example

A small battery may have a nominal voltage of:

1.5 V

UK domestic mains has a nominal voltage of approximately:

230 V AC

The water analogy

Imagine water in a pipe.

Pressure can help drive water through the pipe.

Voltage can similarly help drive current through an electrical circuit.

But remember:

Voltage is not literally water pressure.

It is an analogy.

Easy Glossary

Voltage: The electrical push/difference that can drive current.

Volt: The unit used to measure voltage.

Potential difference: Another technical name for voltage between two points.

MODULE 5 — CURRENT

What is current?

Electric current is the **rate at which electric charge flows**.

It is measured in:

amperes (A)

Usually called:

amps

For example:

A device may draw **2 A** under particular operating conditions.

Easy comparison

Voltage = push

Current = flow rate

Resistance = opposition

Easy Glossary

Current: The rate at which electric charge flows.

Ampere: The unit used to measure electric current.

Amp: Short name for ampere.

MODULE 6 — RESISTANCE

Resistance describes how strongly something opposes electric current.

It is measured in:

ohms (Ω)

A higher resistance generally means less current for a given voltage.

Example

If voltage stays the same:

More resistance → **less current**

Less resistance → **more current**

Easy Glossary

Resistance: Something that makes current harder to flow.

Ohm: The unit used to measure resistance.

MODULE 7 — OHM'S LAW

One of the most important beginner formulas is:

$$V = I \times R$$

Where:

- **V** = voltage
- **I** = current
- **R** = resistance

You can rearrange it:

Current

$$I = V \div R$$

Resistance

$$R = V \div I$$

Example

Voltage:

$$12 \text{ V}$$

Resistance:

$$6 \Omega$$

Therefore:

$$I = 12 \div 6$$

$$I = 2 \text{ A}$$

Remember

Voltage = Current × Resistance

MODULE 8 — POWER AND ENERGY

These two words are often confused.

Power

Power tells us **how quickly energy is being transferred or used**.

Unit:

Watt (W)

Energy

Energy describes the amount transferred or used.

Household electricity consumption is commonly measured in:

kilowatt-hours (kWh)

The difference

Watt = rate

kWh = amount over time

Example:

A:

2 kW heater

running for:

3 hours

uses:

6 kWh

Easy Glossary

Power: How quickly electrical energy is being used or transferred.

Watt: Unit of power.

Energy: The amount of energy transferred or used.

Kilowatt: 1,000 watts.

Kilowatt-hour: A unit of energy equal to using 1 kilowatt for 1 hour.

MODULE 9 — UK ELECTRICITY

UK domestic electricity uses:

230 V nominal AC

and:

50 Hz

The supply reaches homes through a large electricity network.

A simplified journey is:

Generation

↓

Transmission

↓

Distribution



Local supply



Meter



Consumer unit



Household circuits



Appliance



Important

The electricity network uses different voltage levels at different stages.

The very high voltages used for transmission are **not the same voltage delivered to ordinary domestic sockets**.



Easy Glossary

Mains electricity: Electricity supplied from the public electricity network.

Transmission: Moving electricity long distances through the high-voltage network.

Distribution: Delivering electricity through the local network.

Substation: Equipment/site where electrical voltage is transformed and the network is controlled.

Meter: A device that measures electricity consumption.

MODULE 10 — AC AND DC

DC — Direct Current

In the basic model, DC has current flowing in one direction.

Examples include:

- Batteries
- Many electronic circuits
- Solar cells

AC — Alternating Current

AC periodically changes direction.

UK mains electricity is AC.

Memory

DC → Direct

AC → Alternating

Easy Glossary

Direct current: Current that has one direction in the basic model.

Alternating current: Current that repeatedly changes direction.

Frequency: How many cycles happen each second.

Hertz: The unit of frequency.

50 Hz: 50 cycles per second.

MODULE 11 — LIVE, NEUTRAL AND EARTH

A typical modern UK cable may contain three conductors:

Conductor	Typical colour	Basic role
Live	Brown	Carries the supply to the equipment
Neutral	Blue	Normally provides the return path
Earth	Green/yellow	Protective conductor

Live

The live conductor is dangerous.

It is connected to the supply's live potential during normal operation.

Neutral

The neutral conductor normally provides the return path for current in a circuit.

 **Never assume neutral is safe to touch.**

A conductor should not be touched merely because it is called “neutral”.

Earth

Earth is primarily a **protective connection**.

Under fault conditions, it can help provide a path for fault current and allow protective devices to operate.

Key idea

Earth is for protection — it is not simply another neutral.

Easy Glossary

Live: The supply conductor that can present a dangerous voltage.

Neutral: A conductor normally connected to the supply's neutral point and used as part of the circuit return.

Earth: A protective conductor/connection used to help reduce danger during faults.

Fault: A problem that causes an electrical system to behave abnormally.

MODULE 12 — UK PLUGS, SOCKETS AND FUSES

The familiar UK domestic plug is commonly known as:

Type G

It has three pins.

UK plugs are associated with:

BS 1363

The three connections are:

Live — Brown

Neutral — Blue

Earth — Green/yellow

Many UK plugs also contain a fuse.

Why the fuse matters

A fuse is designed to interrupt the circuit when excessive current causes the fuse element to operate.

Its purpose includes helping protect the flexible cable and equipment from certain overcurrent conditions.

Important correction

Do **not** memorise:

“Small appliance = always 3 A.”

Instead:

Use the fuse rating specified or appropriate for the particular appliance/cable, following the manufacturer's instructions and applicable standards.

Common UK plug fuse ratings include **3 A and 13 A**, but the correct choice depends on the appliance and its design.

MODULE 13 — THE CONSUMER UNIT

The **consumer unit** is the main distribution and protection equipment in many homes.

People sometimes call it:

“the fuse box”

Modern consumer units can contain devices such as:

- Main switch
 - Circuit breakers
 - RCDs
 - RCBOs
 - Other protective equipment
-

What does it do?

It distributes electricity into different circuits and incorporates protective devices.

For example, a home may have separate circuits for:

- Lighting
- Socket outlets
- Cooking equipment
- Other fixed equipment

Easy Glossary

Consumer unit: The main electrical distribution and protection equipment in a home.

Circuit breaker: A device designed to automatically interrupt a circuit under specified abnormal current conditions.

RCD: A device designed to disconnect a circuit when it detects a residual current above its operating threshold.

RCBO: A device combining functions associated with overcurrent protection and residual-current protection.

MODULE 14 — ELECTRICAL PROTECTION

Different protective devices deal with different dangers.

Device	Basic purpose
Fuse	Interrupts excessive current
Circuit breaker	Automatically interrupts certain excessive-current/fault conditions
RCD	Detects residual current and disconnects under specified conditions
Earth	Provides protective fault-current path

Insulation

Helps prevent unwanted contact/current paths

Important

It is misleading to say:

“One device protects against everything.”

Electrical safety uses **layers of protection**.

MODULE 15 — ELECTRICAL HAZARDS

Electricity can cause:

Electric shock

Current passing through the body.

Burns

Electrical energy can produce dangerous heat.

Fire

Faults, overheating or unsuitable conditions can cause fire.

Equipment damage

Incorrect electrical conditions can damage equipment.

Warning signs

Watch for:

 Burning smell

 Scorch marks

-  Melted plastic
 -  Damaged cables
 -  Exposed conductors
 -  Unusual buzzing
 -  Sparks
 -  Hot plugs or sockets
 -  Repeated tripping
-

MODULE 16 — SAFE EVERYDAY BEHAVIOUR

Good habits

- Keep electrical equipment away from water unless designed for that environment.
 - Replace damaged equipment.
 - Check cables visually for damage.
 - Avoid inappropriate use of extension leads.
 - Keep heaters clear of materials that could overheat or catch fire.
 - Do not put objects into sockets.
 - Do not bypass protective devices.
 - Follow manufacturer instructions.
 - Know where your consumer unit is.
 - Know how to obtain professional help.
-

Do not

- Touch exposed conductors.
- Work on live mains circuits.
- Ignore repeated protective-device operation.
- Continue using equipment that shows signs of overheating.
- Bypass a fuse or protective device.
- Assume “the switch is off” automatically means a circuit is safe to work on.

MODULE 17 — WHEN SOMETHING GOES WRONG

Scenario A — Damaged cable

You notice exposed conductor on an appliance cable.

Response:

Stop using the equipment.

Do not attempt an improvised repair.

Follow appropriate manufacturer guidance or obtain competent repair/replacement.

Scenario B — Burning smell

You smell burning around a socket.

Response:

Treat it as a potential electrical hazard.

Do not investigate by touching the socket.

If safe to do so, remove the electrical load from use and isolate the relevant supply where appropriate. Seek professional assistance.

If there is immediate danger such as smoke or fire, prioritise emergency safety procedures.

Scenario C — Circuit repeatedly trips

A protective device repeatedly operates.

Don't:

Keep resetting it again and again.

Instead:

Treat repeated operation as a warning that something may be wrong.

Disconnecting a suspected appliance may be appropriate where it can be done safely, but persistent problems require appropriate investigation.

MODULE 18 — EVERYDAY ELECTRICITY

Electrical energy can become different useful outputs.

Device	Main useful output
Lamp	Light
Kettle	Heat
Fan	Movement
Speaker	Sound
Computer	Information processing
Refrigerator	Cooling

 Think:

Electrical energy → device → useful output

MODULE 19 — BASIC CALCULATIONS

Formula 1 — Ohm's Law

$$V = I \times R$$

Formula 2 — Current

$$I = V \div R$$

Example:

$$24 \text{ V} \div 8 \text{ } \Omega = 3 \text{ A}$$

Formula 3 — Power

For a simple DC/resistive calculation:

$$P = V \times I$$

Example:

$$12 \text{ V} \times 2 \text{ A} = 24 \text{ W}$$

Formula 4 — Energy

For a simple constant-power example:

$$\text{Energy} = \text{Power} \times \text{Time}$$

If:

$$2 \text{ kW} \times 3 \text{ hours}$$

then:

$$6 \text{ kWh}$$

MODULE 20 — PRACTICAL OBSERVATION

These are **observation exercises**, not mains electrical work.

Exercise 1 — Electrical vocabulary

Find five electrical items in your home.

For each one, identify:

- What is it?
 - What does it use electricity for?
 - Does it produce heat, light, movement, sound or information?
-

Exercise 2 — Rating label

Look at the rating information on an appliance **without opening it**.

Find:

- Voltage
- Power
- Other ratings shown

Record them.

Exercise 3 — Consumer unit

Locate your consumer unit.

Do not open it.

Identify visually:

- Main switch
- Circuit labels
- Protective devices

The purpose is recognition, not electrical work.

Exercise 4 — Hazard spotting

Look around your home for:

- ☐ Damaged cables
- ☐ Hot plugs
- ☐ Overloaded extensions
- ☐ Burn marks
- ☐ Unusual buzzing
- ☐ Equipment near water
- ☐ Damaged plugs

Do not touch or repair anything dangerous.

09 — PRACTICAL APPLICATION

The Level 1 Decision Method

When you encounter an electrical problem, ask:

STOP

S — See the warning sign

T — Think before touching

O — Observe from a safe position

P — Professional help if necessary

10 — KEY TAKEAWAYS

If you remember only these, remember these:

ELECTRICITY

Electrical systems transfer and use electrical energy.

VOLTAGE

Electrical potential difference that can drive current.

CURRENT

Rate of flow of electric charge.

RESISTANCE

Opposition to current.

POWER

Rate of energy transfer.

ENERGY

Amount transferred or used.

CIRCUIT

A suitable path for electrical operation.

UK BASICS

230 V nominal AC

50 Hz

Brown = Live

Blue = Neutral

Green/yellow = Earth

SAFETY

Fuses, circuit breakers, RCDs, earthing and insulation perform different protective roles.

MOST IMPORTANT

Do not turn Level 1 knowledge into unsafe practical work.

11 — CONCLUSION

Electricity becomes much easier once the vocabulary stops being mysterious.

Remember the chain:

Voltage can drive → current through → resistance → transferring energy → producing power → doing useful work.

Then add the UK safety system:

Live + Neutral + Earth + Protection + Safe behaviour

You do not need to become an electrician to understand electricity.

But if you want to become one, **this foundation is only the beginning.**



EASY GLOSSARY

The Easy Glossary

AC — Alternating Current

Electric current that repeatedly changes direction.

Easy: Electricity that changes direction again and again.

Amp / Ampere

The unit used to measure electric current.

Easy: A number that tells you how much electric charge is flowing.

Appliance

A device that uses electricity to perform a job.

Easy: An electrical thing that helps you do something, such as a kettle.

Circuit

A path that allows an electrical system to operate.

Easy: The route electricity uses to do its job.

Circuit Breaker

A protective device that automatically switches a circuit off under certain abnormal conditions.

Easy: A safety switch that can turn itself off when there is a problem.

Conductor

A material that allows electric charge to move relatively easily.

Easy: A material that lets electricity move through it.

Consumer Unit

The main electrical distribution and protection equipment in a home.

Easy: The important electrical box that controls and protects many home circuits.

Current

The rate at which electric charge flows.

Easy: How much electric charge is moving.

DC — Direct Current

Current that flows in one direction in the basic model.

Easy: Electricity moving mainly in one direction.

Earth

A protective electrical connection/conductor used to help reduce danger during faults.

Easy: A safety path that helps protect people when something goes wrong.

Electric Charge

A property of matter that produces electrical effects.

Easy: A tiny electrical property that can cause electrical forces.

Electron

A tiny negatively charged particle.

Easy: A very tiny particle with a negative electrical charge.

Energy

The ability to cause change or do work.

Easy: Something that can make things happen.

Fault

A problem that makes an electrical system behave incorrectly or unsafely.

Easy: Something has gone wrong.

Frequency

The number of cycles that happen in a given time.

Easy: How many times something repeats.

Fuse

A protective device designed to interrupt a circuit when excessive current causes it to operate.

Easy: A safety part that breaks the electrical path when too much current makes it operate.

Hertz (Hz)

The unit used for frequency.

Easy: A way of counting how many times something repeats each second.

Insulation

Material that helps prevent unwanted electrical contact and current flow.

Easy: A protective covering that helps keep electricity where it should be.

Load

Something that uses electrical energy.

Easy: Something that uses electricity.

Live

The conductor connected to the supply's live potential and capable of causing electric shock.

Easy: The dangerous electricity-carrying wire.

Mains Electricity

Electricity supplied from the public electricity network.

Easy: The electricity that comes into your home from the outside network.

Neutral

A conductor normally connected to the supply's neutral point and used as part of the circuit return.

Easy: The wire normally used to take current back through the circuit.

Ohm (Ω)

The unit of electrical resistance.

Easy: The unit used to describe how much something resists current.

Power

The rate at which energy is transferred or used.

Easy: How quickly energy is being used.

RCD

Residual Current Device.

A protective device that detects residual current and disconnects under specified conditions.

Easy: A fast safety device that switches electricity off when it detects a dangerous imbalance.

Resistance

Opposition to electric current.

Easy: Something that makes it harder for current to flow.

Short Circuit

An unintended low-resistance path that can produce a very large current.

Easy: Electricity has found a dangerous shortcut.

Socket

A fixed outlet designed to accept a compatible plug.

Easy: The place in the wall where a plug goes.

Voltage

Electrical potential difference between two points.

Easy: The electrical push or difference that can drive current.

Volt (V)

The unit used to measure voltage.

Easy: The unit that tells you the voltage number.

Watt (W)

The unit used to measure power.

Easy: The unit that tells you how quickly electrical energy is being used.

Kilowatt (kW)

1,000 watts.

Easy: A thousand watts.

Kilowatt-hour (kWh)

A unit of energy equal to using 1 kilowatt for 1 hour.

Easy: A way of measuring how much electricity has been used over time.

12 — Q&A

Q1. Is electricity the same as electrons?

A: No. Electrons are particles with electric charge. Electricity is a broader term involving electrical charge, current, fields and energy.

Q2. What is voltage?

A: Electrical potential difference.

Q3. What is current?

A: The rate at which electric charge flows.

Q4. What is resistance?

A: Opposition to electric current.

Q5. What is power?

A: The rate at which energy is transferred or used.

Q6. What is a circuit?

A: A suitable electrical path that allows a system to operate.

Q7. What is UK domestic mains voltage?

A: Approximately **230 V AC** nominally.

Q8. What is UK mains frequency?

A: 50 Hz.

Q9. What colour is the modern UK live conductor?

A: Brown.

Q10. What colour is neutral?

A: Blue.

Q11. What colour is earth?

A: Green/yellow.

Q12. Does the earth wire normally carry the load current?

A: No. It is a protective conductor and should not normally carry load current.

Q13. What does a fuse do?

A: It is designed to interrupt excessive current when the fuse operates.

Q14. What does an RCD do?

A: It detects residual current and disconnects under specified conditions, providing an important layer of protection against electric shock.

Q15. Can I assume a circuit is safe because its switch is off?

A: No. Switching off is not, by itself, proof that a circuit is safe to work on.

13 — NORK REVISION CHECK

 **NEW KNOWLEDGE**

Potential new knowledge:

- Electrical charge
 - Electron
 - Voltage
 - Current
 - Resistance
 - Ohm's Law
 - Power
 - Energy
 - AC
 - DC
 - Live
 - Neutral
 - Earth
 - Fuse
 - RCD
 - Consumer unit
-

rNork — Repeat

Repeat these until you can explain them without looking:

Voltage $\neq$ current

Power $\neq$ energy

Earth $\neq$ neutral

Fuse $\neq$ RCD

Switch off $\neq$ proven safe

gNork — Gained Nork

You can mark a topic **gNork** when you can:

1. Define it.
2. Explain it simply.
3. Give an example.
4. Explain why it matters.

pNork — Passed Nork

Aim to pass when you can complete the assessment without relying on the course.

LEVEL 1 TEST

PART A — MULTIPLE CHOICE

1. What unit measures voltage?

- A. Amp
- B. Watt
- C. Volt
- D. Ohm

2. What unit measures current?

- A. Volt
- B. Ampere
- C. Watt
- D. Ohm

3. What unit measures resistance?

- A. Ohm
- B. Watt
- C. Volt
- D. Ampere

4. What does resistance do?

- A. Opposes current
- B. Creates voltage
- C. Measures energy
- D. Stores power

5. UK domestic mains electricity is normally:

- A. DC
- B. AC
- C. Static electricity
- D. Battery electricity

6. What is the nominal UK domestic supply voltage?

- A. 12 V
- B. 110 V
- C. 230 V
- D. 1,000 V

7. What colour is the modern UK live conductor?

- A. Blue
- B. Brown
- C. Green/yellow
- D. White

8. What colour is the earth conductor?

- A. Brown
- B. Blue
- C. Green/yellow
- D. Black

9. What is the main purpose of a fuse?

- A. Store electricity
- B. Increase voltage
- C. Interrupt excessive current when it operates
- D. Create current

10. What does an RCD detect?

- A. Temperature only
- B. Residual current
- C. Light
- D. Sound

PART B — TRUE OR FALSE

11.

A circuit provides a path for electrical operation.

True / False

12.

Power and energy mean exactly the same thing.

True / False

13.

The earth conductor is primarily a protective conductor.

True / False

14.

A damaged electrical cable should continue to be used carefully.

True / False

15.

UK mains frequency is 50 Hz.

True / False

PART C — SHORT ANSWERS

16.

Explain voltage in simple language.

17.

Explain current in simple language.

18.

Explain resistance in simple language.

19.

What is the difference between AC and DC?

20.

Why is electrical insulation important?

PART D — CALCULATIONS

21.

A circuit has:

12 V

and:

6 Ω

Calculate the current.

Formula:

$$I = V \div R$$

22.

A device operates at:

12 V

and:

2 A

Calculate power.

Formula:

$$P = V \times I$$

23.

A 2 kW heater operates for 3 hours.

How much energy does it use?

PART E — SCENARIOS

24.

You discover a cable with exposed copper.

What should you do?

25.

A socket becomes unusually hot and smells of burning.

What should you do?

26.

A circuit breaker repeatedly trips.

Should you simply keep resetting it?

Explain.

PART F — HIGHER-ORDER THINKING

27.

Why does an electrical system need several different safety measures rather than relying on one device?

28.

Explain the difference between:

Voltage → Current → Resistance → Power → Energy

using your own words.



ANSWER KEY

Part A

1. **C — Volt**
 2. **B — Ampere**
 3. **A — Ohm**
 4. **A — Opposes current**
 5. **B — AC**
 6. **C — 230 V**
 7. **B — Brown**
 8. **C — Green/yellow**
 9. **C — Interrupt excessive current when it operates**
 10. **B — Residual current**
-

Part B

11. **True**
12. **False**
13. **True**

14. **False**

15. **True**

Part C — Model Answers

16.

Voltage is the electrical potential difference that can drive current.

17.

Current is the rate at which electric charge flows.

18.

Resistance is opposition to electric current.

19.

AC periodically changes direction; DC flows in one direction in the basic model.

20.

Insulation helps prevent unwanted electrical contact and current paths.

Part D

21.

$$I = V \div R$$

$$I = 12 \div 6$$

$$I = 2 \text{ A}$$

22.

$$P = V \times I$$

$$P = 12 \times 2$$

$$P = 24 \text{ W}$$

23.

$$\text{Energy} = \text{Power} \times \text{Time}$$

$$2 \text{ kW} \times 3 \text{ hours}$$

$$6 \text{ kWh}$$

Part E

24.

Stop using the damaged equipment and do not touch exposed conductors. Obtain appropriate repair/replacement.

25.

Treat it as a potential electrical hazard. Do not touch or investigate the socket. Isolate the supply if this can be done safely and appropriately, and seek professional assistance.

26.

No.

Repeated operation indicates a possible fault, overload or other problem. Repeatedly resetting the device without addressing the cause is unsafe.

Part F — Model Answers

27.

Different protective measures address different hazards. For example, overcurrent protection and residual-current protection detect different types of abnormal conditions, while insulation and earthing provide additional layers of protection.

28.

Voltage is electrical potential difference.

Current is the rate of charge flow.

Resistance opposes current.

Power is the rate of energy transfer.

Energy is the amount transferred or used.



LEVEL 1 COMPLETION CHECK

You have achieved the **Level 1 learning objective** when you can confidently:

- ☒ Explain electricity at a basic level
 - ☒ Explain what a circuit is
 - ☒ Define voltage
 - ☒ Define current
 - ☒ Define resistance
 - ☒ Use Ohm's Law
 - ☒ Explain power and energy
 - ☒ Distinguish AC and DC
 - ☒ Identify UK live, neutral and earth colours
 - ☒ Explain the basic purpose of a fuse
 - ☒ Explain the basic purpose of an RCD
 - ☒ Recognise common electrical hazards
 - ☒ Explain why insulation matters
 - ☒ Recognise when professional help is required
 - ☒ Use the **Easy Glossary** independently
 - ☒ Pass the Level 1 assessment
-



THE FINAL LEVEL 1 MEMORY CARD

ELECTRICITY IN 30 SECONDS

 **Electricity:** electrical phenomena involving charge and energy.

 **Voltage:** electrical potential difference.

 **Current:** rate of charge flow.

 **Resistance:** opposition to current.

 **Power:** rate of energy transfer.

 **Energy:** amount transferred or used.

 **Circuit:** suitable path for electrical operation.

 **UK domestic supply:** approximately 230 V AC, 50 Hz.

 **Live:** Brown.

 **Neutral:** Blue.

 **Earth:** Green/yellow.

 **Protection:** Fuses, circuit breakers, RCDs, earthing and insulation have different jobs.

 **Safety:** If you don't have the competence to do electrical work, don't improvise — get appropriate professional help.

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Update: 19th September 2026